

HARDIK SRIVASTAVA

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[Portfolio](#) | [LinkedIn](#) | [LeetCode](#) | [ResearchGate](#) | [Google Scholar](#) | [GitHub](#)

SKILLS

ML/Research	Sequence/Probabilistic Modeling, Semi-Supervised Learning, Graph-ML, NLP, Ranking, Transformers, xAI
CS/Tooling	C++, Python, SQL, PyTorch, TensorFlow, Caffe, Sklearn, Spark, Kafka, W&B, HuggingFace, Optuna
Infra/MLOps	AWS, Docker, Kubernetes, Flask, MLFlow, NVIDIA Triton, Feature Stores, Data Version Control, Jenkins

EDUCATION

University of Washington	Seattle, WA
<i>Master of Science in Data Science</i>	Sept 2025 - Mar 2027 (Expected)

SRM Institute of Science and Technology	Chennai, TN
<i>B.Tech in Computer Science Engineering with specialization in Big Data Analytics (GPA - 3.82/4)</i>	May 2019 - Jun 2023

EXPERIENCE

Applied Scientist 2	Jun 2023 - Aug 2025
JPMorgan Chase & Co.	Hyderabad, TS

- Led **fraud detection** initiatives saving over **\$220M annually** by developing high-recall classifiers using **XGBoost** and **CatBoost**; leveraged **multi-objective optimization** to preserve feature interpretability for business partners.
- Deployed Semi-Supervised Graph Neural Networks on 60M+ transaction nodes; introduced **topology-aware edge sampling** and **dynamic neighborhood aggregation**, boosting **fraud-detection precision** by **9.2%** over 2M trans/day.
- Designed and productionized a CLIP-style **Contrastive Pre-Trained Transformer** (ViT encoder & BERT-style decoder) for **Document OCR** to detect Check fraud, processing 500K+ checks daily with a **19% boost** in field-level text accuracy.
- Built **Learning-to-Rank** entity matching system for **KYC** using Pairwise Ranking loss to alert fraud entities; Outperformed the TF-IDF baseline on 3B+ records, cutting **62% false positives** and saving **700+** manual review hours daily.
- Developed **data pipelines** for high-volume transactions in PySpark. Performed **optimal feature selection** using **Decision Trees** that boosted **Fraud Capture rate** by **14%** and enhanced predictive accuracy for risk scoring models.
- Collaborated with **JPMC AI Research** on an **internal whitepaper** on **xAI techniques** (SHAP, monotonicity constraints and surrogate tree approximators); **mentored 2 interns** on **cross-lingual NER** and **data drift detection**.

Research Intern	Jun 2022 - Sept 2022
McGill University - <i>Mitacs Scholar</i>	Montreal, QC

- Developed a **Vocabulary Expansion** model for **AAC research** using Affinity Propagation and **SWOW** embeddings. Baseline comparisons (**COCA**) revealed a **4.41 unit mAP gain** and a **21% decrease** in OOV words for test cues.
- Developed **RoBERTa-based sentence embeddings** to capture word-context in image captions. Achieved **Krippendorff's** alpha reliability score of **0.82**; deployed and productionized in **Click AAC** mobile application.

Machine Learning Engineer - Intern	Jan 2022 - Jun 2022
Samsung Research	Bangalore, KN

- Optimized **word-context relationships** for Bixby search using bipartite graphs, reducing **false-positive** preds by **12%**. Matched **FP32** perf for INT8 inference using quantization-aware training for word-disambiguation task with **2.9%** F1 gain.

PROJECTS

Auto-Explainability and Evaluation of LLMs Developed a framework for Transformer explainability by visualizing prediction attributions using Integrated Gradients in TensorFlow. Integrated with MLFlow Evaluation API for auto-evaluation.

No-Code Algo-Trading Engine Developed a domain-specific language with Mean Reversion strategies to create & execute trades. Added a risk-assessment model using Backpropagation, reducing risk by 15.2% compared to open-sourced models.

PUBLICATIONS

- [Srivastava, H.](#) (2023). Multi-Modal Sentiment Analysis Using Text and Audio for Customer Support Centers
- [Srivastava, H.](#) (2023). Neural Text Style Transfer with Custom Language Styles for Personalized Communication Systems
- [Srivastava, H.](#) (2021). Using NLP Techniques for Enhancing Augmentative and Alternative Communication Applications

ACCOMPLISHMENTS

- **Certifications** — AWS Certified Cloud Practitioner, AWS Certified Machine Learning Specialist
- **1st Position** — JPMC ScaleUp (2024), JPMC AWS DeepRacer Regionals (2023), Smart India Hackathon Nationals (2023)